

Kinematic Mirror Mount

Command Reference Manual

1 Overview

The mount has two independent axes, each a brushed DC motor with an analog Hall position sensor:

- **Axis 1** — motor M1, sensor a0
- **Axis 2** — motor M2, sensor a1

Commands are plain text, one per line, and are identical over both transports. In the syntax below, **N** is the axis number (1 or 2) and sensor values are raw ADC counts (0–4095).

Transport	Details
USB serial	115200 baud, newline line endings. Always available; the only transport that accepts provisioning commands.
WiFi (TCP)	Port 3333, discovered via mDNS as <code>mirrormount-<id>.local</code> . Requires authentication (Section 3).

Responses are text lines, mirrored to USB *and* the connected WiFi client.

2 First-time setup (over USB)

Connect the mount by USB and open a serial console (e.g. `pio device monitor`). Then:

1. Set the device login password:
`setpass <password>`
2. Give it your WiFi network:
`wifi <ssid> <password>`
3. It confirms and prints its address, e.g.
`wifi: connected ip=192.168.1.42 name=mirrormount-ab12.local`

Both are stored in flash and reconnect automatically on power-up. There are **two separate secrets**: the *login password* (`setpass`, authenticates clients) and the *WiFi credentials* (`wifi`, joins your network).

Command	Description
<code>setpass <password></code>	Set the login password (stored hashed). USB only.
<code>wifi <ssid> <password></code>	Save WiFi credentials and connect. USB only. Password may contain spaces.
<code>wifi</code>	Show connectivity status (SSID, IP, name, signal).
<code>factory-reset</code>	Clear WiFi + password (keeps axis calibration). USB only.

3 WiFi authentication

On each TCP connection the device issues a challenge; the client proves it knows the password without sending it:

```
device -> NONCE <hex>                                (random, per connection)
client -> AUTH <hex>      hex = HMAC-SHA256(K, nonce)
device -> OK                                (authenticated) -- or -- DENIED
```

where `K = SHA256(password)`. After `OK`, the same commands as the serial console are accepted. A Python client (interactive sessions over WiFi or USB, plus a `provision` helper) is available in the companion repository `physiverse_monolithic_motorized_mirror_mount_client`; after `pip install .` it provides the `mount` command.

4 Control commands

Accepted on both transports (WiFi requires authentication first).

Command	Description
mN <speed> [ms]	Drive motor N at speed (−255 to 255; sign = direction). Optional ms auto-stops after that long.
m <a> [ms]	Drive both motors at once.
seekN <target> <speed> [stall_ms]	Closed-loop move until sensor = target . speed is a magnitude; direction is automatic. Clamped to the calibrated range; gives up after stall_ms of no progress (default 2500).
homeN [speed]	Drive axis N to its home position (saved home, else range midpoint).
home	Home <i>both</i> axes simultaneously.
jogN <speed> [max_ms]	Bounded manual jog while streaming the sensor (default cap 2000 ms; any key aborts on USB).
probeN <speed>	Step-and-settle travel characterization; reports the edges and learns polarity.
findN <speed>	Drive toward an end of travel while streaming position; any key stops it (USB), and the stop position is reported as a range MIN/MAX candidate for setrangeN .
tickN <±n> [spd] [ms]	Fine adjustment: one continuous drive worth n minimal ticks (sign of n = direction). Pulse size defaults to the per-axis saved values (speed 100 / 5 ms unless changed).
settickN <spd> <ms>	Save axis N’s tick pulse size to flash.
setrangeN <min> <max>	Save axis N usable travel range to flash.
sethomeN <pos>	Save a custom home position (pos < 0 clears → range midpoint).
setpolN <+1 −1 0>	Override / clear the learned motor-to-sensor polarity.
s	Stop both motors.
r [n]	Read a0/a1, averaged over n samples (default 1).
status	Report both axes: speed, polarity, range, home.
?	Help.

5 Notes

- **Calibration** (polarity, range, home) is per-axis, stored in flash, and survives reboot and factory-reset.
- **seek/home/timed** moves are self-terminating (range clamp + stall guard). If the WiFi client disconnects mid-move, they finish.
- An open-ended **mN <speed>** (no duration) runs until **s**; if the authenticated WiFi client is lost, this is the one motion that is auto-stopped (fail-safe).
- The sensor is non-linear near its travel extremes (“soft shoulders”); the calibrated range excludes them, and seek targets are clamped into it.
- The motors require power on the VIN screw terminal (5–12 V); USB alone powers only the logic.

6 Typical session

```
# over USB, once:
setpass labpass123
wifi MyNetwork mypassword

# then, from a client over WiFi (or over USB):
status
home
seek1 2300 150
seek2 2100 150
tick1 -5          ; small nudge, axis 1, toward lower sensor values
r 16
```